

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

6th Semester of B. Pharm. Examination

Theory University Examination October 2017

PH323 Medicinal Chemistry-II

Date: 02.10.2017 , Monday Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

Q 1	Attempt all questions. Each question is of one mark.	20
1.	chlorthalidone is chemically _____ derivative [A] quinazolinone [B] indoline [C] 1- oxoisindiline [D] thiazide	1
2.	Which of the followings is Benzothiazepine class of drug? [A] Bepridil [B] verapamil [C] nifedipine [D] Diltiazem	1
3.	Which of the following derivatives of barbitures is inactive? [A] 1,3- disubstituted barbituric acid [B] 5,5' disubstituted barbituric acid [C] 5,5'- disubstituted thio barbituric acid [D] 1,5,5'- trisubstituted barbituric acid	1
4.	Benzocain is a _____ [A] Alkyl ester of Amino benzoic acid [B] Amino amide [C] Amino ketone [D] Amino ether	1
5.	Which of the following is atypical antipsychotic agents? [A] Fluphenazine [B] Olanzapine [C] Thioridazine [D] Aripiprazole	1
6.	If 6 Member central ring of phenothizine ring is replace by 7 member ring then activity is change to [A] Antipsychotic [B] Antidepressant [C] Anxiolytic [D] Anticonvulsant	1

7.	Class IA anti arrhythmic agents [A] shortens action potential duration [B] no effect on action potential duration [C] increase threshold of action potential. [D] prolongs action potential duration	1
8.	Flecainide is [A] Class IC anti arrhythmic agent [B] Class IA anti arrhythmic agent [C] Class IB anti arrhythmic agent [D] Class II anti arrhythmic agent	1
9.	C-19 skeleton in steroid is a representation of [A] gonane [B] androstane [C] estrane [D] Pregnane	1
10.	maximum antidepressant activity is obtained when amino group separated from tricyclic nucleus by [A] ethylene bridge [B] butylene bridge [C] propylene bridge [D] Large alkyl chain	1
11.	substituent which increase glucocorticoid activity is [A] 16-alpha-hydroxy [B] 16-alpha-methyl [C] 16-beta-methyl [D] 6-alpha-fluoro	1
12.	the symbol delta (Δ) is used in steroid nomenclature for indicating [A] missing methyl group [B] double bond [C] position of hydroxyl group [D] stereochemistry of steroid	1
13.	Iodide in thyroid synthesis is activated by [A] thyroglobuline [B] tyrosine [C] peroxidase [D] lysosomal proteolysis	1
14.	Which of the following is selective serotonin reuptake inhibitor? [A] phenelzine [B] iproniazide [C] lithium carbonate [D] fluoxetine	1
15.	Which of the following is fibrate derivative with antihyperlipidemic activity? [A] Gemfibrozil [B] Colestipol [C] Ezetimib [D] Probucol	1
16.	Which one of the following is 4-anilinopiperidines analogue? [A] Pentazocine [B] Methadone [C] Naloxone [D] Fentanyl	1
17.	1,4-Dihydropyridine scaffold possessing Calcium Channel antagonist is... [A] Verapamil	1

	[B] Diltiazem [C] Nifedipine [D] Bepridil	
18.	Milrinone, a cardiotonic agent contains [A] Piperazine ring [B] Pyridine ring [C] Pyrimidine ring [D] Pyrrole ring	1
19.	One of the following structural feature makes thiazide diuretics weak acidic in their nature [A] Sulfonamide [B] Carboxylic acid [C] Halogen [D] Trifluoromethyl	1
20.	Aspirin can be prepared by the reaction between [A] Salicylic acid and acetic anhydride [B] p-amino phenol and acetic anhydride [C] Anthranilic acid and acetic anhydride [D] Benzoic acid and acetic anhydride	1
SECTION – II		
Q 2	Attempt any TWO of the following	
A	Describe SAR of barbiturates.	05
B	Write a detail note on chemistry of tricyclic anti depressants.	05
C	Enlist various drugs used as an anti parkinsonism with their structures.	05
Q 3	Attempt any TWO of the following	
A	Give chemical classification of antihypertensive agents with atleast one structure from it.	05
B	Classify drugs used as NSAIDs with their structure (atleast one structure from each class).	05
C	Draw a synthetic pathway for lidocaine and Ibuprofen.	05
SECTION – III		
Q 4	Attempt any FOUR of the following	
A	Explain SAR of Thiazide diuretics.	05
B	Write a short note on CNS stimulants.	05
C	Discuss SAR of Calcium Channel Blockers.	05
D	Outline the biosynthesis of thyroid hormones.	05
E	Write a chemistry of glucocorticoids.	05
F	Write a short note on oral hypoglycemic agents.	05
Q 5	Attempt any FOUR of the following	
A	Explain SAR of HMG Co-A Reductase inhibitors in details.	05
B	Explain SAR and therapeutic application of eicosanoids.	05
C	Give chemical classification of drugs used as antiarrhythmic agents.	05
D	Write a chemistry of benzodiazepines.	05
E	Write a SAR of ACE inhibitors.	05
F	Explain chemistry of morphine analogues.	05